

Tia Rice

708-506-2694 | tiarice2028@u.northwestern.edu | [linkedin.com/in/tia-rice](https://www.linkedin.com/in/tia-rice) | github.com/Compsci-RiceTia

EDUCATION

Northwestern University

Evanston, IL

B.S. in Electrical Engineering, Minor in Artificial Intelligence

Expected 2028

- **Relevant Coursework:** Probabilistic Systems, Signals and Systems

EXPERIENCE

Computer Engineering Undergraduate Research Assistant

Jun. 2025 – Present

Northwestern University

Evanston, IL

- Developing **PyTorch**-based autonomous agents for multi-agent patrolling on TurtleBot3 robots using **ROS2**
- Evaluating and improving model performance through comparative testing

Media and Design Studio Lab Aide

Oct. 2024 – Present

Northwestern University

Evanston, IL

- Open the studio for daily operation while ensuring equipment and workspaces are ready for users
- Manage equipment checkout and troubleshoot basic computer hardware and software issues

Physics Undergraduate Research Assistant

Sep. 2023 – Jun. 2025

Northwestern University | Fermilab

Evanston, IL

- Collaborated on a Python-based simulation of dark matter interactions in scintillating bubble chambers
- Used **Ubuntu** and **Anaconda** to manage software packages, install libraries, and configure development environments
- Co-authored and presented a publication based on the project's findings
- Presenting research to research team and at a formal academic conference

PROJECTS

Radio-Controlled Mini Baja SAE Car

Present

- Designing the chassis, suspension, and powertrain in SolidWorks
- Developing Arduino test programs to validate electronic speed controller, servo, and radio receiver functionality

Design Thinking & Communication Quarter Project

Dec. 2024

- Worked in a four-person team over 10 weeks with a \$100 budget to design, CAD, and prototype a portable grocery assistance device
- Applied human-centered design principles to develop a lightweight, socially acceptable solution tailored to client needs

TECHNICAL SKILLS

Programming: Python, C/C++, Java, C#, MATLAB, Arduino, RTL/Verilog

Tools: Git, Linux, Docker, Microsoft Office

Design Software: SolidWorks, KiCad

Manufacturing: Lathe, Standard Mill, Conversational Mill, Bandsaw

Languages: English, Spanish, German

ACTIVITIES

Baja SAE

Present

Northwestern University

Evanston, IL

- Serving as primary lathe manufacturer for suspension, chassis, powertrain, ergonomics, and electronics
- Design a PCB for wheel control and develop fuel-sensing circuitry for real-time refueling feedback

National Society of Black Engineers (NSBE)

Present

General Member

Evanston, IL

- Volunteer at the Evanston Public Library to introduce children to engineering through hands-on cardboard carnival game projects